

Instructor Notes:



Instructor Notes:**Document History**

Date	Course Version No.	Software Version No.	Developer / SME	Change Record Remarks
21-Jul-2006	Ver1.0	1.2.6	Shrilata Tavargeri	NA
Jan-2009	Ver2.0	2.5	Shrilata Tavargeri	Changed material with new changes introduced in ver 2.5 and inputs from BU
Aug-2011	Ver 3.0	3.0	Shrilata Tavargeri	Changed material with new changes introduced in ver 3.0 and inputs from BU
June 2013	Ver 4.0	3.0	Mohan Chinnaiah	Revamped materials according to new requirements
May 2015	Ver 5.0	4.0	Rathnajothe Perumalsamy	Changed material with new changes introduced in ver 4.0
June-2016	Ver 6.0	4.0	Vinod Satpute Yukti Valecha Tanmaya Acharya	Modified as per Toc for ELTP
Feb-2018	Ver 7.0	5.0	Bharati Thorat	Modified as per TOC of ELTP

Capgemini

Keep this as a hidden slide.

Note to co-ordinators: Not to be printed for the class book.

Instructor Notes:

Document History				
Date	Course Version No.	Software Version No.	Developer / SME	Change Record Remarks
21-Jul-2020	Ver8.0	5.0	Vaishali Srivastava	NA

Capgemini

Keep this as a hidden slide.

Note to co-ordinators: Not to be printed for the class book.

Instructor Notes:

Course Goals and Non Goals



- **Course Goals**

- Understand the benefits of using Spring
- Understand the principles of IoC
- Understand the Java Based Configuration
- Connect business objects to persistent stores using Spring's DAO modules
- Use the Spring MVC web framework to develop flexible web applications
- Use the Spring MVC Java Based Configuration to develop flexible web application

- **Course Non Goals**

- Design patterns, Spring Integration with different technologies



Instructor Notes:

Pre-requisites



- Core Java , Java 8 features and JDBC
- XML, DBMS/SQL
- Servlets, JSP
- Concepts of MVC, Design patterns

Instructor Notes:

Intended Audience

- All Java application developers especially
Enterprise Java Programmers
- Software designers



Instructor Notes:

Day Wise Schedule



Day 1

- Lesson 1: Introduction to Spring Platform and environment, Spring Boot features
- Lesson 2: Introduction to Spring Framework, IoC
- Lesson 3 : Spring Java Base Configuration

Day 2 [Online]

- [Spring Boot 1 \[Online\]](#)
- [Spring Boot 2\[Online\]](#)

Day 3

- [Spring RESTful\[Online\]](#)

Instructor Notes:

Day Wise Schedule



Day 4

- [Spring Data JPA\[Online\]](#)
- Lesson 6 : Spring Data JPA Integration

Day 5

- [Spring Data Rest\[Online\]](#)

Day 6

- [Swagger and Spring Fox\[Online\]](#)
- Spring MCQ 4 Test: 40 Qs

Instructor Notes:

Table of Contents



Lesson 1: Introduction to Spring Platform and Environment

- Introduction to Spring Platform and environment
- Spring 5 features
- Spring Projects At a Glance
- Spring IO Platform
 - Spring Framework
 - Spring Boot

Lesson 2: Introduction to Spring Framework, IoC

- What is Spring Framework, Benefits of Spring
- The Spring architecture
- Dependency Injection
- IOC – Inversion of control, wiring beans
- Bean containers, lifecycle of beans in containers, Bean Scopes
- Customizing beans with Post Processors
- Annotation-based configuration

Instructor Notes:

Table of Contents



Lesson 3 : Spring Java Base Configuration

- Spring Application Using Java Base Configuration
- Java Configuration Class
 - @Configuration
 - @Bean
- Implementing Bean Lifecycle Callbacks and Bean Scope
- Registering Configuration Using AnnotationConfigApplicationContext

Instructor Notes:

Table of Contents



Lesson 4: Spring Boot

Lesson 5: Spring RESTful

Lesson 6: Spring JPA Integration

- Spring support for JPA
- Implementing Spring JPA integration
- Spring Data JPA

Instructor Notes:

References



- Spring in Action, Fourth Edition, Manning publications by Craig Walls
- Spring-framework-reference.pdf from SpringSource (this is available in the downloaded Spring software)



Instructor Notes:

Software required



- JDK version 1.8 + with help, Netscape or IE
- MS-Access/Connectivity to Oracle database
- WildFly
- Eclipse Luna with Spring Tool Suite
- Spring 4.0 API with docs

Instructor Notes:

Other Parallel Technology Areas



- EJB 3.0
- PicoContainer
- NanoContainer
- Keel Framework
- Google Guice

Capgemini 

- PicoContainer: is an exceptionally small DI (Dependency Injection) container that allows to use DI for your application without introducing any dependencies other than PicoContainer itself
- NanoContainer: is an extension to PicoContainer for managing trees of individual PicoContainer containers.
- Keel Framework: is more of a metaframework, in that most of its abilities come from other frameworks that are all brought together under one roof.
- Google Guice: focuses purely on DI.